Indroneel Roy

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Education

Shahjalal University of Science and Technology (SUST) February 2020 - August 2025
B.Sc. in Electrical & Electronic Engineering ***GPA: 3.22/4.00*** *(WES Evaluation)*

Technical Skills

Languages: Python, SQL, C, Matlab, Assembly (Intel ×86)
ML/DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV
Computer Vision: CNNs, Object Detection (YOLO), Image Segmentation, ResNet, U-Net
LLM & NLP: LangChain, LangGraph, Transformers, Hugging Face, RAG, Vector Databases
MLOps & Deployment: Docker, FastAPI, Streamlit, Hugging Face Spaces, Git, AWS
Databases: SQL, Vector Databases (Pinecone, Chroma)
Algorithms & DS: Data Structures, Algorithms (200+ LeetCode problems solved)

Projects

HybridCNN-Polyp-Segmentation | ***PyTorch, OpenCV*** GitHub

- Design and implement a hybrid CNN-Transformer architecture integrating ResNet-50 encoder with multi-head self-attention mechanisms
- Develop novel cross-attention fusion module to synergistically combine local CNN features with global Transformer representations
- Leverage skip connections and hierarchical decoding for precise polyp boundary delineation across multiple scales
- Reduce computational complexity while maintaining high accuracy

Vision-Based-Wheat-Detection | ***PyTorch, OpenCV*** GitHub

- Trains a YOLOv11-Medium model on the wheat detection dataset
- Achieved **94.7% mAP@50** and **55.4% mAP@50–95**, with **91.7% precision** and **89.2% recall** on **29,422+ annotated instances**.

Agentflow-Chatbot Multi-Tool Conversational AI | ***LangChain, LangGraph, LLM*** GitHub

- Design and implement stateful conversational AI system using LangGraph state machine architecture with persistent SQLite checkpointing for multi-turn dialogue management
- Develop comprehensive tool integration framework incorporating 10+ specialized functions including web search (DuckDuckGo), financial APIs (Alpha Vantage, CoinGecko), weather services, and real-time data retrieval
- Leverage LangChain and Groq LLM (Llama-3.3-70B) with dynamic tool binding and conditional edge routing to enable autonomous function calling and context-aware responses

Cassava Leaf Disease Classification using Vision Transformer (ViT) | ***PyTorch*** GitHub

- Designed and implemented a Vision Transformer (ViT-Base) model for multi-class plant disease classification, achieving **87.15%** validation accuracy across **5 disease categories** on the Kaggle Cassava Leaf Disease dataset.
- Leverage transfer learning with timm library and PyTorch framework, optimizing 224×224 RGB image processing pipeline with GPU acceleration for production deployment
- Achieved robust classification performance with **93.4%** precision and **95.6%** recall for Cassava Mosaic Disease (CMD), demonstrating strong generalization across bacterial blight, brown streak, green mottle, and healthy leaf classes.

Certifications

Neural Networks and Deep Learning – Coursera (View Certificate)

Research & Model Implementations

- **CNN Architectures** – Implemented and trained LeNet, AlexNet, ResNet, DenseNet, and EfficientNet from scratch in PyTorch; performed architectural comparisons and benchmarked performance on image classification tasks
- **Transformer-Based Models** – Implemented Transformer, Swin Transformer, and TransUNet architectures; explored self-attention mechanisms and hierarchical vision transformers for classification and segmentation
- **Image Segmentation Models** – Implemented U-Net, SegFormer, and MaskFormer; conducted experiments on medical and natural image segmentation datasets using Dice and IoU metrics
- **Code Repository** – All implementations and experiments are publicly available on GitHub

Additional Information

Languages: English (Fluent), Bengali (Native)

Interests: Open-source AI contributions, research paper implementations, technical blogging, cooking, playing chess